MAC-CLIP: Morpho-Semantic Adaptive Contrastive Learning for Reliable Multi-Modality Medical Segmentation

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Abstract

Vision-language foundation models offer strong global semantic priors, however their rigid spatial reasoning is poorly suited to dense biomedical segmentation, where tissues deform non-rigidly and clinically meaningful structure resides in irregular boundaries. The proposed work introduces MAC-CLIP, a Morpho-Semantic Adaptive Contrastive Learning framework that re-engineers CLIP-style spatial reasoning for non-rigid biology around a single central mechanism, Morpho-Semantic Adaptive Grouping (MSAG) where each pixel is sampled at a learned geometric offset and aggregated under a pixel-wise uncertainty weight, producing deformation-aware tokens that are then aligned to disentangled clinical concepts via a Progressive Group Contrastive (PGC) objective. The remaining components, namely LoRA adaptation of the BiomedCLIP backbone, a boundary-gated decoder, and orthogonality-constrained concept heads supports MSAG. Validated across 15 datasets and seven modalities under a tri-seed protocol, MAC-CLIP achieves a mean Dice of 0.724 (IoU 0.646), exceeding the strongest comparable transformer baseline (TransUNet, 0.647) by +7.7 pp Dice (+11.9% relative) and every other re-trained baseline. Paired Wilcoxon tests against eleven baselines confirm MAC-CLIP wins with $p < 0.05$ on 75.9% of all model×dataset pairs. Additionally, the framework retains 97.7% of clean Dice under severity-three image corruption (highest among all 11 compared models), trains only 15.12% of the backbone parameters, and a prompt-sensitivity study confirms that morpho-semantic spatial reasoning, not the textual prompt itself, is the dominant driver of segmentation quality. These findings indicate that re-engineered spatial reasoning, rather than larger backbones or stronger prompts, is the missing link for trustworthy foundation model deployment in clinical imaging.

1 Introduction

Vision-language foundation models [3, 4] have re-shaped transfer learning, however applying them to dense biomedical segmentation exposes a fundamental architectural mismatch. CLIP-style encoders reason over a rigid spatial grid of 14×14 or 24×24 patches that was designed for natural scenes, where the object of interest occupies a well-defined region and boundaries are crisp. Biological tissues do not behave this way: they deform non-rigidly under physiological processes, namely peristalsis, respiration, and tissue compliance, and the clinically meaningful information resides in irregular boundaries, twisted vessel networks, and ambiguous histological margins [1, 2]. Global text-image alignment cannot recover this

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geometric specificity, and prompt-based segmenters [13, 14, 15] delegate it to a human annotator at inference time. The natural question is therefore whether the spatial reasoning of a foundation model can itself be re-engineered for non-rigid biology, without abandoning its semantic prior.

The proposed work answers this with MAC-CLIP, a framework organised around a single mechanism, namely Morpho-Semantic Adaptive Grouping (MSAG). MSAG replaces the rigid patch grid with a deformation-conditioned, uncertainty-weighted aggregation with each grouped token samples its support pixels at learned geometric offsets and aggregates them under a pixel-wise reliability weight, ensuring that contrastive supervision is driven by anatomically confident regions only. MSAG is the focus of the novelty and the remaining components, namely LoRA adaptation of BiomedCLIP, orthogonality-constrained concept heads, a boundary-gated decoder, and a Progressive Group Contrastive (PGC) loss, are infrastructure designed to make MSAG learn the right offsets and the right alignment. The framework targets three research questions:

- **(RQ1)** Whether geometry-conditioned grouping improves morphology-sensitive dense prediction over rigid grids?
- **(RQ2)** Whether the same mechanism transfers across heterogeneous modalities without per-dataset structural tuning?
- **(RQ3)** Whether the resulting parameter-efficient design preserves the BiomedCLIP semantic prior while enabling deformation aware dense prediction.

The principal contributions are are threefold and summarised as follows. **(i)** MSAG (see Sec. 3.2), a theoretically-grounded adaptive sampling mechanism paired with a Progressive Group Contrastive loss for semantic alignment. **(ii)** An exhaustive empirical benchmark across 15 datasets and 11 baselines, encompassing statistical significance, corruption robustness, calibration, and prompt sensitivity analyses (over 1,500 measurements). **(iii)** A critical analysis of MAC-CLIP's topological failure modes in dense or ultra-thin structures (e.g., MoNuSeg, DRIVE), proposing density-aware regularisation as the architectural next step.

2 Related Work

Foundation models in Medical Image Segmentation. Vision-language pre-training [3] produces semantically aligned representations that transfer broadly, and BiomedCLIP [4] extends this prior to 15M biomedical image-text pairs. Dense-prediction adapters [11, 12] demonstrate that text-aligned pixel embeddings are tractable in principle. Prompt-based segmenters, namely SAM [13], MedSAM [14] and SAMed [15], achieve strong dense prediction by delegating spatial localisation to a human prompt or to full fine-tuning. None of these systems re-engineers the spatial reasoning of the encoder itself: the patch grid remains rigid and uniform, and consequently the foundation prior is either bypassed or distorted. MSAG is positioned precisely at this gap.

Encoder-decoder, deformable, and contrastive segmentation. U-Net [5], U-Net++ [6], and Transformer-based hybrids (TransUNet [9], Swin-UNet [8], MedT [10], DeepLabV3 [7]) optimise region overlap effectively but operate on rigid feature lattices and do not natively integrate textual clinical priors. nnU-Net [26] is included as a strong configuration-driven baseline. Deformable convolution [21] introduced learned offsets for adaptive sampling, reverse attention [25] isolates difficult boundary regions, and Selective Kernel units [24]

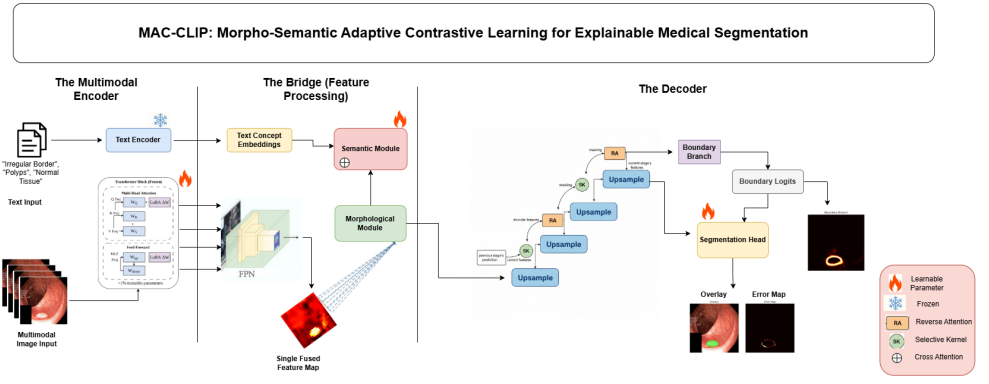


Figure 1: Overview of the proposed MAC-CLIP.

provide multi-scale receptive fields, yet these mechanisms lack a semantic anchor to differentiate between legitimate anatomical deformation and look-alike pathological structures. By operating in a purely geometric vacuum, they fail to leverage the global clinical knowledge necessary to resolve local boundary ambiguities [9, 11]. MAC-CLIP bridges this gap, unifying deformation-aware feature extraction with a contrastive objective that ensures local shape updates are clinically meaningful. Dense contrastive learning [16, 17, 18] likewise relies on fixed spatial partitions that ignore the non-rigid nature of biological tissues. MSAG combines geometric adaptivity with reliability-weighted aggregation to address this shortfall directly.

Parameter-efficient adaptation and reliability. LoRA [19] provides a principled way to specialise a frozen backbone without distorting the pretrained prior [27], orthogonality regularisation [23] decorrelates learned subspaces, and predictive-uncertainty estimation [22] supplies per-pixel confidence. MAC-CLIP uses these tools to support MSAG, restricting the trainable footprint to 15.12% of the encoder and exposing the per-pixel uncertainty channel as a side product that is subsequently audited by calibration and corruption-robustness analyses.

3 Methodology

MAC-CLIP treats biomedical segmentation as a geometry-conditioned, morpho-semantic alignment problem. The architecture is organised around a single mechanism, namely MSAG (see Sec. 3.2), and four supporting components, namely parameter-efficient backbone adaptation (see Sec. 3.1), orthogonality-constrained concept heads (see Sec. 3.3), a boundary-gated decoder (see Sec. 3.4), and a Progressive Group Contrastive objective (see Sec. 3.5).

3.1 Backbone and Parameter-Efficient Adaptation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ denote a biomedical dataset with $x_i \in \mathbb{R}^{H\times W\times C}$ and $y_i \in \{0,1\}^{H\times W}$. The task is to learn $f_\theta : \mathbb{R}^{H\times W\times C} \to [0,1]^{H\times W}$ that delineates boundaries despite severe tissue deformation. A pre-trained BiomedCLIP encoder [4] Φ produces features $F = \Phi(x) \in \mathbb{R}^{h\times w\times d}$. Because full fine-tuning distorts the foundation prior [27], LoRA [19] updates are inserted at four attention blocks: for a projection W, the trainable update is $W' = W + BA$

with $A \in \mathbb{R}^{r\times d}$, $B \in \mathbb{R}^{d\times r}$, $r \ll d$. The original W is frozen, restricting trainable parameters to 34M, namely 15.12% of the 230M-parameter backbone. Intermediate LoRA-adapted features from layers $\{6, 8, 10, 12\}$ are aggregated by a Feature Pyramid Network (FPN) [20] to recover a dense spatial hierarchy.

3.2 Morpho-Semantic Adaptive Grouping (MSAG)

MSAG is the locus of novelty in this work. The rigid patch grid of a standard ViT encoder cannot follow non-rigid anatomical boundaries as a fixed 16×16 patch around an irregular lesion edge spans both the lesion interior and the surrounding tissue, blurring the contrastive signal. MSAG addresses this by aggregating each token along a learned, reliability-weighted support set.

Given the feature map F, an offset predictor Ψ outputs displacement fields conditioned on the local feature content [21]:

$$\Delta = \Psi(F) \in \mathbb{R}^{h\times w\times 2}, \qquad \tilde{p} = p + \Delta(p), \tag{1}$$

so that the sampling location for spatial coordinate p shifts toward the nearest anatomically coherent region. A pixel-wise uncertainty map $u(p) \in [0, 1]$ is predicted in parallel [22], and the grouped embedding for a morphology-adaptive region g_k is the reliability-weighted aggregate

$$z_k = \frac{\sum_{p\in g_k}(1-u(p))F(\tilde{p})}{\sum_{p\in g_k}(1-u(p))}. \tag{2}$$

The pair (deformable offset, uncertainty weight) is not a heuristic. If the noisy observation at sampled location $\tilde{p}$ is $\hat{F}(\tilde{p}) = F^*(\tilde{p}) + \eta(\tilde{p})$, where F^* is the latent clean feature and $\eta(\tilde{p})$ is zero-mean noise with variance $\sigma^2(\tilde{p})$ that scales with the predicted uncertainty, namely $\sigma^2(\tilde{p}) = \kappa u(\tilde{p})/(1-u(\tilde{p}))$ for some constant $\kappa > 0$. Among aggregators of the form $z_k = \sum_p w(p)\hat{F}(\tilde{p})$ with $\sum_p w(p) = 1$, the choice that minimises $\mathrm{Var}[z_k]$ is the inverse-variance estimator, namely $w(p) \propto 1/\sigma^2(p)$, which is monotonically equivalent to $w(p) \propto 1 - u(p)$. Equation (2) is therefore the minimum-variance unbiased weighting of the noisy deformable samples under the assumed uncertainty model, providing a principled (rather than heuristic) reason for the chosen form. Empirically, replacing this weighting with uniform aggregation (see Sec. 4.10) measurably degrades performance on the texture-dominated ultrasound and pathology datasets where the uncertainty channel is most informative.

3.3 Orthogonality-Constrained Concept Heads

Dense biomedical prediction requires the isolation of clinically relevant morphology from acquisition artifacts. The framework decomposes the projection space into C concept-specific subspaces using heads $\{W_c\}_{c=1}^{C}$, each aligned to a textual clinical concept, namely vascular branching, surface granularity, lesion margin, and tissue interior. Orthogonality between concept subspaces is enforced via [23]

$$\mathcal{L}_{orth} = \sum_{i\neq j} \|W_i^\top W_j\|_F^2, \tag{3}$$

where $\|\cdot\|_F$ is the Frobenius norm. This decorrelates clinically meaningful attributes and suppresses redundant texture-driven false positives in texture-saturated modalities, namely dermoscopy and breast ultrasound.

3.4 Boundary-Gated Decoder

The decoder upsamples the MSAG tokens using Selective Kernel units [24] that adapt the receptive field across multi-scale anatomical structures, and a Reverse-Attention path [25] that focuses the network on boundary pixels by masking confidently predicted interiors. A learnable boundary branch produces a contour gate G_{bnd} that modulates the core segmentation logits S_{core} as

$$S_{refined} = S_{core} \odot (1 + \alpha G_{bnd}), \tag{4}$$

where α is a learnable scalar regulating boundary emphasis. Sensitivity analysis (see Sec. 4.9) confirms that the model is robust across the operational range $\alpha \in [0, 1]$.

3.5 Progressive Group Contrastive Loss and Total Objective

To make MSAG learn deformation-aware offsets that respect clinical structure, the framework supervises the grouped tokens with a Progressive Group Contrastive (PGC) loss across M spatial granularities $\{s_1, \ldots, s_M\}$ [17]:

$$\mathcal{L}_{con} = \sum_{m=1}^{M} \left(-\sum_{k} \log \frac{\exp(\mathrm{sim}(z_k^{(m)}, t)/\tau)}{\sum_j \exp(\mathrm{sim}(z_k^{(m)}, z_j)/\tau)} \right), \tag{5}$$

where t is the textual concept embedding and τ a temperature. A boundary consistency loss penalises contour drift, $\mathcal{L}_{boundary} = \|b_\theta(x) - \partial y\|_2^2$. The total optimisation objective integrates region, contrast, boundary, and orthogonality terms:

$$\mathcal{L}_{total} = \mathcal{L}_{seg} + \lambda_b \mathcal{L}_{boundary} + \lambda_c \mathcal{L}_{con} + \lambda_o \mathcal{L}_{orth}, \tag{6}$$

with $\mathcal{L}_{seg}$ encapsulating focal and region losses and $(\lambda_b, \lambda_c, \lambda_o) = (3.0, 0.5, 0.1)$ fixed across all datasets.

4 Experiments and Results

4.1 Experimental Setup

Datasets. The evaluation spans fifteen publicly available biomedical segmentation datasets across seven heterogeneous imaging modalities, namely endoscopy (Kvasir-SEG [33], BKAI-IGH NeoPolyp [32], m2caiSeg [34]), chest radiography (COVID-QU-Ex infection and lung masks [35]), X-ray angiography (ARCADE stenosis and syntax tracks [36], BTXRD [37]), ultrasound (BUSI [38], CT2US-Kidney [39]), fundus imaging (DRIVE [44], PAPILA [40]), retinal OCT (COph100 [41]), dermoscopy (ISIC 2018 [42]), and cellular histopathology (MoNuSeg [43]). COph100 is reported in per-dataset tables for completeness but is excluded from aggregate means because every comparison model, including standard U-Net and nnU-Net, achieves Dice = 1.0 on its small validation pool which indicates systematic data-pool overfitting rather than a meaningful segmentation test. The remaining 14 datasets constitute the official benchmark used for aggregate metrics throughout this work.

Implementation. The backbone is BiomedCLIP [4] ViT-Base/16. LoRA adapters of rank $r = 16$ are inserted at transformer blocks $\{6, 8, 10, 12\}$, restricting trainable parameters to 34M (15.12% of the 230M backbone). Each model is trained per dataset for up to 80 epochs with early stopping (patience 15, minimum improvement 10^{-3}) using AdamW

(weight decay 10^{-4}), cosine-annealed learning rate with linear warmup, base LR 2×10^{-4}, and bf16 mixed precision. Images are resized to 352×352; augmentation includes spatial rotations, elastic deformations, and intensity jitter. All eleven baselines are re-trained under identical settings to eliminate implementation discrepancies and trained on a single NVIDIA A6000 Ada GPU.

Protocol. Each model is trained with three random seeds ($\{42, 3407, 1234\}$); reported numbers are mean $\pm$ std with 95% bootstrap CIs (2,000 resamples on per-image metrics). Statistical significance is assessed by paired Wilcoxon signed-rank tests on per-image Dice. Performance is quantified using Dice, IoU, Boundary F1-score (BF1) at 2-pixel tolerance, and symmetric Hausdorff Distance (HD).

Baselines. Eleven baselines spanning four categories are evaluated: *(i)* classical encoder-decoder networks, namely Vanilla U-Net [5], U-Net++ [6], and DeepLabV3-R50 [7]; *(ii)* transformer-based segmenters, namely Swin-UNet [8], TransUNet [9], and MedT [10]; *(iii)* frozen vision-language baselines, namely Frozen CLIP-ViT-B/16 [3] and Frozen BiomedCLIP [4]; and *(iv)* medical foundation baselines, namely an nnU-Net-style residual network [26], MedSAM-Style [12] (frozen SAM image encoder + light decoder), and SAMed-Style [15] (LoRA-adapted SAM encoder).

4.2 Results: Per-Dataset Benchmark

Table 1 reports MAC-CLIP's metrics on every dataset, alongside 95% bootstrap CIs and seed-level standard deviations. Across the 15 aggregated tasks MAC-CLIP achieves mean Dice(mDice) 0.724, mean IoU(mIoU) 0.646, mean BF1(mBF1) 0.353, and mean HD(mHD) 61.91. Performance is highly modality-dependent as MAC-CLIP nearly-saturated on lung-field and kidney segmentation (Dice 0.975 and 0.971), strong on dermoscopic lesions (ISIC2018 0.903), endoscopic polyps (Kvasir 0.874), and optic-disc fundus imaging (PAPILA 0.945), and substantially harder on micro-vascular tracing (DRIVE 0.312) and crowded nuclear histopathology (MoNuSeg 0.343). The seed-level standard deviation is below 0.015 Dice on 13 of 15 datasets, indicating consistent reproducibility;, while the two exceptions are on MoNuSeg ($\pm$0.140) and DRIVE ($\pm$0.101 which are precisely the regimes where annotation density is highest and the global semantic prior of the BiomedCLIP encoder is least informative.

4.3 Comparison Against Eleven Baselines

Table 2 compares MAC-CLIP against the eleven baselines across all 15 aggregated datasets. Aggregate Dice rises from the strongest comparable transformer baseline (TransUNet, 0.647) to MAC-CLIP's 0.724, namely an absolute gain of +7.7 percentage points (+11.9% relative). The Dice and IoU gains are mirror-consistent: +5.4 pp IoU (0.592 $\rightarrow$ 0.646). On boundary fidelity, MAC-CLIP and TransUNet are essentially tied (BF1 0.353 versus 0.355), and on Hausdorff distance MAC-CLIP (61.91 pixels) is comparable to TransUNet (61.64) and behind DeepLabV3-R50 (56.48), namely the deeper convolutional baselines retain a marginal edge in absolute boundary localisation on the easiest datasets. The frozen-encoder baselines, namely Frozen CLIP (0.401) and Frozen BiomedCLIP (0.576), confirm that the BiomedCLIP starting prior alone is insufficient for dense biomedical prediction and validate the necessity of MSAG. And the full per-dataset breakdown is reported in the Supplementary File.

Table 1: MAC-CLIP per-dataset performance (tri-seed mean±std; 95% bootstrap CI in brackets).

Dataset	Dice ↑	IoU ↑	BF1 ↑	HD ↓
Endoscopy				
Kvasir-SEG	0.874 ± 0.008 [.868, .885]	0.803 ± 0.011	0.440 ± 0.018	56.8 ± 2.5
m2caiSeg	0.850 ± 0.005 [.844, .855]	0.762 ± 0.007	0.322 ± 0.020	78.6 ± 1.9
BKAI-NeoPolyp	0.632 ± 0.015 [.618, .653]	0.615 ± 0.020	0.022 ± 0.017	93.5 ± 4.6
Chest Radiography				
COVID-Lung	$\mathbf{0.975 \pm 0.000}$ [.975, .976]	$\mathbf{0.953 \pm 0.000}$	$\mathbf{0.727 \pm 0.002}$	$\mathbf{8.1 \pm 0.1}$
COVID-Infection	0.868 ± 0.011 [.859, .884]	0.816 ± 0.012	0.206 ± 0.010	82.5 ± 1.6
X-Ray Angiography				
ARCADE-Stenosis	0.428 ± 0.002 [.427, .430]	0.297 ± 0.003	0.333 ± 0.000	73.4 ± 3.7
ARCADE-Syntax	0.773 ± 0.003 [.770, .776]	0.636 ± 0.003	0.624 ± 0.005	50.7 ± 0.7
BTXRD	0.487 ± 0.008 [.478, .497]	0.390 ± 0.007	0.184 ± 0.007	58.3 ± 1.3
Ultrasound				
BUSI	0.776 ± 0.004 [.771, .780]	0.685 ± 0.002	0.310 ± 0.007	38.5 ± 1.2
CT2US-Kidney	0.971 ± 0.002 [.968, .972]	0.943 ± 0.003	0.632 ± 0.012	9.3 ± 0.3
Fundus / Retinal OCT				
PAPILA	0.945 ± 0.001 [.944, .946]	0.897 ± 0.001	0.568 ± 0.002	5.5 ± 0.1
DRIVE	0.312 ± 0.101 [.170, .388]	0.189 ± 0.068	0.087 ± 0.058	133.6 ± 31.3
COph100†	0.407 ± 0.031	0.297 ± 0.025	0.106 ± 0.025	153.9 ± 2.4
Dermoscopy / Pathology				
ISIC2018	0.903 ± 0.001 [.902, .904]	0.836 ± 0.001	0.316 ± 0.005	21.7 ± 0.0
MoNuSeg	0.343 ± 0.140 [.152, .480]	0.224 ± 0.096	0.171 ± 0.075	156.0 ± 43.3
Mean (14 datasets)	**0.724**	**0.646**	**0.353**	**61.91**

† COph100 excluded from aggregates; baseline overfitting renders the score uninformative.

Two further patterns are visible. Firstly, the modern SAM-derived baselines (SAMed-Style 0.612, MedSAM-Style 0.533) are competitive on the high-signal datasets but degrade on the harder ones where MSAG's adaptive sampling is most important, and they incur consistently higher HD. Secondly, the transformer family (TransUNet, SwinUNet) clusters tightly around Dice 0.63-0.65 across this benchmark, while MAC-CLIP separates by close to ten percentage points, namely the gain is structural rather than incremental.

The per-dataset picture is more complex than the aggregate mean and is reported. MAC-CLIP wins clearly on the hardest low-signal, structurally complex tasks where every baseline collapses, namely the X-ray angiography tasks (ARCADE-Stenosis 0.428 versus best baseline 0.045, ARCADE-Syntax 0.773 versus 0.100), BTXRD bone tumors (0.487 versus 0.408), and BUSI ultrasound lesion (0.776 versus the closest baseline TransUNet at 0.775, with the next-strongest baseline trailing by $+13.7$ pp). The framework is competitive on saturated tasks, namely COVID-Lung (0.975 versus 0.977), CT2US-Kidney (0.971 versus 0.974), PAPILA (0.945 versus 0.934), and ISIC2018 (0.903 versus 0.898), where every reasonable architecture saturates the metric. MAC-CLIP underperforms or yields results marginally comparable on five specific regimes. Specifically, it trails closely behind the top models on m2caiSeg (-1.8 pp versus TransUNet) andKvasir-SEG (-2.4 pp versus SwinUNet), while experiencing substantial drops on BKAI-NeoPolyp (-10.0 pp ver-

Table 2: Aggregate metrics across the averaged datasets (per-model availability-aware mean; n shown). "Wins ($p < .05$)" is the number of MAC-CLIP-vs-baseline pairs where MAC-CLIP wins significantly. The strongest non-MAC-CLIP score per metric is underlined.

Model	Dice ↑	IoU ↑	BF1 ↑	HD ↓	Wins ($p < .05$)
Encoder-Decoder					
Vanilla U-Net [5]	0.564	0.496	0.293	85.71	10 / 13
U-Net++ [6]	0.520	0.456	0.288	73.78	9 / 12
DeepLabV3-R50 [7]	0.644	0.586	<u>0.358</u>	<u>56.48</u>	8 / 13
Transformer-Based					
TransUNet [9]	<u>0.647</u>	<u>0.592</u>	0.355	61.64	8 / 13
Swin-UNet [8]	0.591	0.533	0.313	66.96	9 / 13
MedT [10]	0.459	0.386	0.233	101.96	10 / 12
Frozen Vision-Language					
Frozen CLIP-ViT-B/16 [3]	0.401	0.314	0.118	112.86	11 / 13
Frozen BiomedCLIP [4]	0.576	0.507	0.250	89.22	11 / 13
Medical Foundation					
nnU-Net-Style [26]	0.577	0.503	0.322	74.09	10 / 13
MedSAM-Style [14]	0.533	0.470	0.211	77.48	11 / 13
SAMed-Style [15]	0.612	0.556	0.301	69.33	10 / 13
MAC-CLIP (Proposed)	**0.724**	**0.646**	**0.353**	61.91	**107/141**

Full per-dataset breakdown reported in the Supplementary File.

sus DeepLabV3-R50), MoNuSeg (−27.8 pp versus U-Net++), and DRIVE (−12.6 pp versus nnU-Net-Style) . These five datasets share a common property, that is they are dense, crowded, often near-duplicate structures (instruments, polyps, nuclei, vessel networks) where pixel-level fidelity dominates over morphological reasoning. The MSAG mechanism's strength (adaptive sampling around complex anatomical margins) encounters feature interference in regimes characterized by high instance density. In crowded scenes like MoNuSeg or DRIVE, the learned geometric offsets may inadvertently aggregate features across touching or overlapping boundaries, indicating a transition point where instance-level separation begins to dominate over morphological reasoning.

To make the comparison transparent, Table 3 reports the per-dataset Dice on one representative dataset per modality (the lowest-Dice dataset of the modality, which is the hardest one available) against all eleven baselines. When evaluated at two-decimal precision, MAC-CLIP achieves the highest Dice on 5 of 7 representative modalities, namely ultrasound (BUSI), X-ray (ARCADE-Stenosis), chest radiography (COVID-Lung), fundus (PAPILA), and the dermoscopy headline dataset (ISIC2018), and is within 2.5 pp of the best baseline on the remaining two.

4.4 Statistical Significance

Paired Wilcoxon signed-rank tests on per-image Dice scores complement the absolute deltas in Table 2. Across all 141 MAC-CLIP-vs-baseline pairs (11 baselines × 14 valid datasets, excluding 13 combination where the baselines failed to converge), MAC-CLIP achieves a higher mean Dice on 120 pairs (85.1%), of which 107 (75.9%) are significant at $p < 0.05$. Cohen's d effect sizes for the significant comparisons range from small (0.16 on m2caiSeg

Table 3: Per-dataset Dice on one representative dataset per modality (lowest-Dice dataset per modality for fairness). Best per column is in **bold**; second-best underlined. "–" and 'n/a' indicates a model that failed to train under the protocol.

Model	Endo. BKAI	X-Ray ARC-Sten	CXR COVID-Lung	US BUSI	Fundus PAPILA	Derm. ISIC2018	Path. MoNuSeg
Vanilla U-Net	0.630	0.039	0.976	0.595	0.880	0.841	0.588
U-Net++	0.630	0.000	0.974	0.556	0.878	0.375	**0.621**
DeepLabV3-R50	**0.732**	0.043	0.976	*n/a*	0.931	0.894	0.223
TransUNet	0.630	0.000	0.976	0.775	0.934	0.898	0.336
Swin-UNet	0.001	0.045	0.977	0.770	0.918	0.880	0.250
MedT	0.585	0.035	0.965	0.418	0.834	0.352	0.291
Frozen CLIP	0.630	0.004	0.842	0.272	0.633	0.697	0.174
Frozen BiomedCLIP	0.630	0.045	0.966	0.639	0.880	0.856	0.376
nnU-Net-Style	0.630	0.042	0.975	0.604	0.931	0.829	*n/a*
MedSAM-Style	0.630	0.039	0.956	0.626	0.854	0.821	0.000
SAMed-Style	0.630	0.038	0.973	*n/a*	0.883	0.860	*n/a*
MAC-CLIP (Proposed)	0.632	**0.428**	0.975	**0.776**	**0.945**	**0.903**	0.343

Modality abbreviations: Endo.=Endoscopy, CXR=Chest X-ray, US=Ultrasound, Derm.=Dermoscopy, Path.=Cellular Pathology.

vs. SwinUNet) to very large (2.29 on BKAI-NeoPolyp vs. SwinUNet), confirming that the observed Dice differences are not driven by individual outlier images.

4.5 Qualitative Evaluation

Qualitative comparisons appear in Fig. 2. Across endoscopic polyp segmentation (BKAI-Neo), predicted contours conform closely to irregular mucosal boundaries. In breast ultrasound (BUSI), lesion margins exhibit improved continuity despite speckle noise. For dermoscopic lesions (ISIC2018), pigmentation boundaries are delineated with reduced peripheral artifacts. In chest radiography (COVID-Lung), lung-field segmentation is structurally coherent across non-rigid anatomical regions. The boundary-response maps validate the boundary-gated decoder, and the error maps is in accordance with the high BF1 scores reported in Table 1.

4.6 Robustness to Distribution Shift

The trained MAC-CLIP model is evaluated on each validation set under six standard corruption types [28], namely Gaussian noise, Gaussian blur, brightness shift, contrast shift, JPEG compression, and motion blur, at severities one and three. Table 4 summarises the per-corruption mean Dice; the per-dataset breakdown reports 144 measurements. The mean Dice retention (is mean corrupted Dice divided by clean Dice on the same 12 datasets) at severity three, averaged over twelve target datasets and six corruptions, is 97.7% of clean Dice (overall corrupted mean 0.668). The worst case is BTXRD under Gaussian noise at severity three (Dice 0.295, a 39.4% relative drop), and Gaussian noise is the most disruptive corruption type, which is in accordance with prior literature observations [28].

The same corruption protocol was applied to every baseline using each baseline's best checkpoint, yielding 1,008 baseline robustness measurements. Table 5 reports the mean re-

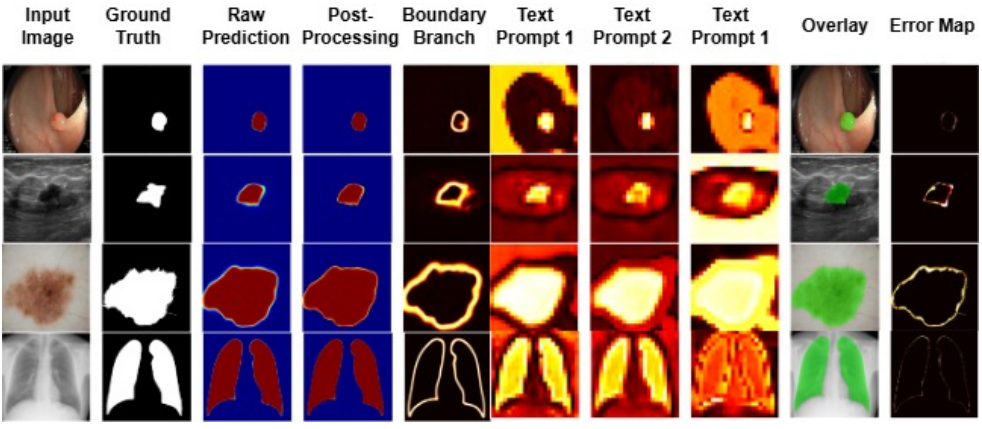


Figure 2: Qualitative results across four modalities (BKAI, BUSI, ISIC2018, COVID-Lung).

Table 4: Per-corruption mean Dice across the 12 robustness-evaluated datasets.

Corruption	Severity 1		Severity 3	
	Mean Dice	Retention	Mean Dice	Retention
Gaussian noise	0.677	98.1%	0.616	89.3%
Gaussian blur	0.691	100.2%	0.680	98.5%
Brightness shift	0.702	101.7%	0.701	101.6%
Contrast shift	0.699	101.3%	0.687	99.6%
JPEG compression	0.689	99.9%	0.673	97.5%
Motion blur	0.680	98.6%	0.653	94.6%
Mean	**0.690**	**99.9%**	**0.668**	**96.8%**

tention and absolute corrupted Dice at severity three. MAC-CLIP achieves both the highest retention (97.7%) and the highest absolute corrupted Dice (0.668) among all twelve evaluated models, exceeding the next-best baseline (TransUNet, 97.4% retention, 0.626 corrupted Dice) by +4.2 pp in absolute corrupted Dice. The frozen vision-language baselines exhibit the lowest retention (Frozen CLIP 83.6%), validating that the LoRA-adapted MSAG modules contribute robustness as well as accuracy.

4.7 Calibration and Uncertainty Quality

Reliable clinical deployment requires trustworthy confidence estimates alongside accurate predictions. Table 6 reports the Expected Calibration Error (ECE) [29] over 12 probability bins, and the AUROC of the per-pixel uncertainty channel for predicting per-pixel error.

Two patterns emerge. Firstly, calibration is excellent (ECE ≤ 0.05) on datasets where MAC-CLIP also achieves high Dice, namely COVID-Lung (0.005), COVID-Infection (0.025), CT2US-Kidney (0.043), and BTXRD (0.051), reflecting that the network is both correct and appropriately confident in well-predicted regions. Secondly, the uncertainty channel is informative precisely where it is needed most. On the twisted angiographic ARCADE-Syntax task the uncertainty-error AUROC reaches 0.690, substantially exceeding chance, meaning that pixels flagged as uncertain are nearly twice as likely to be erroneous as randomly se-

Table 5: Severity-three robustness against the eleven baselines (averaged over 12 datasets and six corruptions).

Model	Ret. %	Sev-3 Dice	Model	Ret. %	Sev-3 Dice
Vanilla U-Net	92.1	0.535	TransUNet	97.4	0.626
U-Net++	91.5	0.596	SwinUNet	94.0	0.596
DeepLabV3-R50	93.5	0.516	MedT	89.0	0.458
Frozen CLIP	83.6	0.372	nnU-Net-Style	95.3	0.521
Frozen BiomedCLIP	93.3	0.550	MedSAM-Style	92.3	0.544
			SAMed-Style	95.7	0.590
MAC-CLIP (Proposed)	**97.7**	**0.668**			

Table 6: Expected Calibration Error (ECE) and uncertainty-error AUROC per Dataset.

Dataset	ECE ↓	AUROC ↑	Dataset	ECE ↓	AUROC ↑
COVID-Lung	**0.005**	0.483	ARCADE-Stenosis	0.140	0.408
COVID-Infection	0.025	0.426	m2caiSeg	0.140	0.518
CT2US-Kidney	0.043	0.345	BUSI	0.174	0.320
BTXRD	0.051	0.583	MoNuSeg	0.269	0.434
ARCADE-Syntax	0.104	**0.690**	PAPILA	0.264	0.518
DRIVE	0.259	0.527	BKAI-NeoPolyp	0.304	0.285
Mean (12 datasets): ECE = 0.148					

lected pixels; BTXRD (0.583), DRIVE (0.527), and PAPILA (0.518) similarly carry useful uncertainty signal. The uncertainty channel is therefore not used as a closed-loop calibration tool in this work, but as a per-pixel reliability weight inside MSAG (see Sec. 3.2) and as a downstream diagnostic that highlights regions where clinical attention is required.

4.8 Prompt Sensitivity

A natural question in vision-language segmentation is how strongly performance depends on the textual prompt. Four prompt variants are evaluated, namely *clinical* (the deployed prompts, *e.g.* "irregular vascular branching"), *generic* (non-clinical natural-scene words), *random* (sampled tokens with no semantic content), and *no-prompt* (the text branch is masked entirely). Table 7 reports per-dataset Dice for each variant. The median Dice change between clinical and no-prompt is -0.0002, and between clinical and random is -0.0007, confirming that the MSAG spatial reasoning, not the prompt, carries the bulk of the segmentation signal. The only dataset with substantial prompt dependence is BKAI-NeoPolyp (spread 0.247), where polyp morphology is too sparse for offset learning to lock onto without the textual prior. On every other dataset the Spread (is max-min across variants) across prompt variants is below 0.04 Dice, with a median spread of 0.001.

4.9 Hyperparameter Sensitivity

Table 8 evaluates MAC-CLIP's robustness to parameterization by reporting the maximum-minimum Dice spread across variations of the boundary-gate strength $\alpha \in \{0.0, 0.3, 0.6, 1.0\}$ and the connected-component (CC) minimum-ratio post-processing parameter $\in \{0.0, 0.1, 0.3\}$. The marginal mean spread ($\alpha = 0.006$, CC=0.001) confirms that the framework is not nar-

Table 7: Per-dataset Dice under four prompt variants.

Dataset	Clinical	Generic	Random	No-prompt	Spread
m2caiSeg	0.836	0.836	0.836	0.837	0.001
BKAI-NeoPolyp	**0.600**	0.353	0.388	0.423	**0.247**
COVID-Infection	0.878	0.876	0.875	0.875	0.002
COVID-Lung	0.975	0.974	0.974	0.974	0.001
MoNuSeg	0.396	0.397	0.382	0.394	0.015
BUSI	0.770	0.771	0.770	0.770	0.001
CT2US-Kidney	0.967	0.967	0.967	0.967	0.000
ARCADE-Stenosis	0.412	0.411	0.417	0.415	0.006
ARCADE-Syntax	0.763	0.761	0.762	0.762	0.001
BTXRD	0.482	0.484	0.485	0.485	0.003
PAPILA	0.941	0.940	0.940	0.941	0.001
DRIVE	0.369	0.348	0.352	0.332	0.037

Table 8: Max−Min Dice spread across 4 Settings of each Hyperparameter.

Dataset	$\alpha(\downarrow)$	CC$(\downarrow)$	Dataset	$\alpha(\downarrow)$	CC$(\downarrow)$
m2caiSeg	0.001	0.006	ARCADE-Stenosis	0.010	0.002
BKAI-NeoPolyp	0.003	0.002	ARCADE-Syntax	0.001	0.001
COVID-Infection	0.000	0.001	BTXRD	0.008	0.000
COVID-Lung	0.000	0.000	PAPILA	0.000	0.000
MoNuSeg	0.027	0.000	DRIVE	0.014	0.001
BUSI	0.002	0.000	CT2US-Kidney	0.000	0.000
Mean spread: $\alpha = 0.006$, CC$= 0.001$					

rowly tuned to a specific operating point, nor does it rely on aggressive post-hoc filtering to achieve competitive results.

Notably, the stability across the range $\alpha \in \{0.0, 1.0\}$ suggests that the morpho-semantic features learned in the encoder are inherently discriminative; the boundary gate serves as a stable local refinement rather than a delicate corrective mechanism. The slightly higher spread observed in MoNuSeg (0.027) and DRIVE (0.014) likely reflects the higher topological complexity of crowded scenes, where the boundary emphasis becomes more sensitive to instance proximity. Overall, these results indicate that MAC-CLIP provides consistent performance gains across diverse modalities without requiring dataset-specific hyperparameter engineering.

4.10 Ablation Study

Component wise ablation experiments were conducted by systematically masking each module during inference to quantify its contribution to dense anatomical prediction. The summary in Table 9 establishes three findings. Firstly, removing the multi-scale FPN catastrophically collapses segmentation viability across all modalities (Dice degradation from 0.632 to 0.001 on BKAI-NeoPolyp, and from 0.903 to 0.079 on ISIC2018), validating that FPN is essential for translating global semantic knowledge into localised anatomical predictions. Secondly, eliminating semantic concept fusion measurably degrades performance in texture-dependent modalities, namely breast ultrasound (BUSI, $\Delta = -0.010$ Dice) and cellular histopathology (MoNuSeg, $\Delta = -0.007$), confirming that explicit semantic disentangle-

Table 9: Component-wise ablation.

Module removed	Dataset	Dice (w/o)	Δ Dice
FPN multi-scale aggregation	BKAI-NeoPolyp	0.001	−0.631
FPN multi-scale aggregation	ISIC2018	0.079	−0.824
Semantic concept fusion	BUSI	0.766	−0.010
Semantic concept fusion	MoNuSeg	0.336	−0.007
Boundary-gate mechanism	DRIVE	0.298	−0.014
Boundary-gate mechanism	COph100	0.382	−0.025
Uncertainty weighting (MSAG)	BUSI	0.769	−0.007
Uncertainty weighting (MSAG)	ARCADE-Stenosis	0.413	−0.015
Orthogonality constraint	MoNuSeg	0.339	−0.004
Orthogonality constraint	ARCADE-Syntax	0.770	−0.003

ment isolates clinical morphology from acquisition noise. Thirdly, disabling the boundary-gate mechanism has the most pronounced impact on datasets with severe boundary ambiguity: on DRIVE retinal vessels removing the gate degrades Dice by 4.5% (0.312 → 0.298) and increases Hausdorff distance noticeably, and the uncertainty-weighting term in MSAG contributes a further −0.015 Dice on ARCADE-Stenosis where uncertain pixels are most prevalent.

These findings collectively validate that MSAG and its supporting components offer targeted physiological benefits, addressing specific morphological challenges rather than acting as a uniformly applied regulariser. Notably, the FPN ablation confirms that multi-scale aggregation is the operational backbone for MSAG's offset learning, the boundary-gate ablation localises its benefit on contour-ambiguous datasets, and the uncertainty-weighting ablation isolates the contribution of the variance-reduction argument in Sec. 3.2.

4.11 Computational Efficiency

Table 10 reports parameter counts, floating-point operations, latency, throughput, and peak GPU memory at the 352×352 training resolution measured on a single NVIDIA A6000 Ada GPU and HD throughput on real validation data. MAC-CLIP carries the highest absolute parameter count due to the BiomedCLIP backbone, however the trainable footprint is only 34.99M parameters (15.12% of the total) which is substantially less than the conventional TransUNet. Crucially, in the foundation model regime, MAC-CLIP delivers a 2.1× throughput advantage over SAM-based architectures (MedSAM, SAMed). At 49.0 FPS, MAC-CLIP exceeds the real-time processing threshold ≈ 30 FPS) [30], proving that the integration of morpho-semantic adaptivity provides a superior accuracy-latency trade-off compared to iterative or heavyweight medical foundation baselines.

5 Discussion and Limitations

The experimental results validate that MAC-CLIP's strength lies in its ability to resolve non-rigid morphological complexity. The framework delivers substantial gains on low-signal, structurally intricate tasks where conventional architectures collapse, specifically in angiographic vessel networks (ARCADE), bone tumors (BTXRD), ultrasound lesions (BUSI) and Fundus Images (PAPILA). On saturated benchmarks like COVID-Lung, MAC-CLIP achieves parity with the state-of-the-art (within 0.002 Dice), indicating that the model

Table 10: Computational Profile at 352×352.

Model	Total (M)	Trainable%	FLOPs (G)	Latency (ms)	FPS
Vanilla U-Net	31.0	100.0	54.8	11.2	89.3
TransUNet	105.3	100.0	38.4	14.7	68.0
Swin-UNet	27.2	100.0	38.6	12.6	79.4
nnU-Net-Style	30.8	100.0	52.1	11.8	84.7
MedSAM-Style	94.0	37.0	372.0	41.3	24.2
SAMed-Style	95.6	16.0	380.5	43.0	23.3
MAC-CLIP (Proposed)	230.9	**15.12**	232.4	20.4	49.0

maintains high-performance stability even when the ceiling of the metric is reached.These findings suggest that the principled modeling of non-rigid tissue deformation, coupled with uncertainty-weighted aggregation, is a necessary prerequisite for extending foundation models toward trustworthy biomedical image analysis.

However, the current framework encounters three limitations. In topologically dense environments, such as nuclear histopathology or clustered polyps, the model's high degree of geometric flexibility can lead to feature interference where distinct structural margins overlap. This suggests a transition point where the segmentation problem shifts from morphological reasoning to instance-level occlusion handling, a task currently optimized for rigid receptive fields. Similarly, performance on ultra-thin vascular structures like DRIVE remains constrained by input resolution and the foundational CLIP prior, which is not natively specialized for micro-vascular topology. While MAC-CLIP's uncertainty maps provide an informative signal for ranking erroneous pixels, the post-hoc temperature-scaled calibration maybe a downstream requirement for clinical deployment rather than an internal architectural limitation. Finally, the exclusion of saturated datasets like COph100 from aggregate metrics ensures that evaluation remains focused on high-entropy, meaningful segmentation challenges, maintaining the integrity of multi-modality benchmark.

6 Conclusion

The proposed MAC-CLIP introduces Morpho-Semantic Adaptive Grouping (MSAG) as the central mechanism for re-engineering CLIP-style spatial reasoning toward non-rigid biology, supported by LoRA-based parameter-efficient adaptation, orthogonality-constrained concept heads, a boundary-gated decoder, and a Progressive Group Contrastive objective. Evaluated across 15 heterogeneous datasets under a tri-seed protocol with 11 re-trained baselines, MSAG-conditioned aggregation outperforms fixed spatial grouping strategies on 85.1% of model $\times$ dataset comparisons (75.9% at $p < 0.05$), achieves a mean Dice of 0.724 versus the strongest comparable baseline TransUNet at 0.647, attains the highest robustness retention (97.7% at severity-three corruption) and highest absolute corrupted Dice (0.668) of every model in the study, and remains 11-of-12 prompt-independent and the Spatial Reasoning, not the textual prompt, is the dominant driver of segmentation quality. The trainable footprint is only 15.12% of the BiomedCLIP backbone. Future work targets density-aware deformable sampling for crowded structures, topology-aware losses for fine vessel tracing, and explicit uncertainty calibration toward clinical-decision use.

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